

Financial planning software does not usually show the implied returns, but Exhibit 3.9 does just that through a reverse-engineering process. With the lump-sum investment, the compounded real return at the twenty-fifth percentile is 3.8 percent over thirty years. For an accumulator, the twenty-fifth percentile return is 3.7 percent, while it is 3.4 percent for the retiree. At the tenth percentile, realized compounded real returns were 2.7 percent for the lump-sum investment, 2.4 percent for the accumulator, and 2.1 percent for the retiree. These numbers are naturally lower to provide a greater chance of success, and sequence-of-returns risk pushes these numbers even lower for accumulators and retirees than with the lump-sum investment that does not experience sequence risk. The volatility of outcomes increases as we transition from a lump sum (standard deviation of 1.9 percent) to accumulation (2.1 percent) to retirement (2.5 percent).

Financial planning software based on Monte Carlo simulations generally presents its results in terms of a probability for success. For instance, retirees may aim for a 90 percent rate of success or higher. An implied rate of return on the portfolio is connected to a given probability of success, though Monte Carlo simulations generally do not express their output in this way. Higher rates of success would be connected with lower portfolio returns, since this return hurdle must be exceeded by the portfolio for the financial plan to be successful. In this discussion, I am tackling Monte Carlo from a different direction—using Monte Carlo simulations first to get a rate of return for the portfolio, then to simulate a financial plan using a fixed rate of return. Conservative investors will want to work with lower assumed returns, implying a need to save more today. The exhibit provides insight about appropriately conservative adjustments for return assumptions.

Individuals accumulating or spending assets will have different experiences than someone using a lump-sum investment. Accumulation effectively places greater importance on the returns earned late in your career when a given return impacts more years of contributions. This is sequence-of-returns risk as it applies in the accumulation phase. With greater importance placed on a shorter sequence of returns, we should expect a wider distribution of outcomes.